Arrays and selectors: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01

The row-minus-column voltages are (1,1)=1.2 V, (1,2)=0.6 V, (2,1)=0.6 V and (2,2)=0 V. The two unselected cells on selected lines are half-selected. Safety depends on pulse duration, repetition and disturbance thresholds, not voltage alone.

E02

The selected path current is $I_s = 1V \times 0.1\,\mu S = 0.1\,\mu A$. Three 1 MΩ cells form a series sneak path: $I_p = 1V/(3M\Omega) = 1/3\,\mu A$. Total sensed current is $0.433\,\mu A$. Each path resistance drops 1/3 V: floating column 2 is at 2/3 V and row 2 at 1/3 V. This applies to the stated floating, linear model only.

E03

$I(0.6V) = 10\,\mu A/10^3 = 10\,nA$. Two bias points do not specify other paths, states, wire drops, sensing circuits or distributions; the ratio alone does not determine array read margin.

E04

The total resistance is 6 kΩ, giving $1.2V/6k\Omega = 0.2mA$. The selector drops 0.4 V and the memory 0.8 V, summing to 1.2 V. For nonlinear devices solve their I–V relations together with the series voltage constraint instead of using fixed-resistance division.

E05

1R has no separate access element; 1T1R controls a series transistor gate; 1S1R uses selector nonlinearity to reduce unselected current; 1D1R uses diode rectification. Both bias directions need sufficient drive, acceptable leakage and voltage distribution, so the name alone does not establish bipolar compatibility.

E06

$4F^2 = 4(20nm)^2 = 1600nm^2$. The ideal three-layer projected area is $1600/3 \approx 533.3nm^2$ per bit. Decoders and sense circuits, interlayer wiring and isolation, leakage, thermal constraints, defects and yield reduce realized chip density.

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Arrays and selectors: Worked solutions (continued)

E07

For the stated LRS and timing, $NL = 10\,\mu A/(10\,nA) = 1000$. The 99 fixed half-selected paths contribute $99(10\,nA) = 0.99\,\mu A$. Adding the selected current gives $I_H = 0.2 + 0.99 = 1.19\,\mu A$ and $I_L = 10 + 0.99 = 10.99\,\mu A$. These totals require the stated clamped node, branch voltages and unchanged branch states. In another topology, node voltages and paths must be solved; NL by itself does not provide them or the HRS response.

E08

If all 100 pass independently, the multiplication rule gives $P(\text{all pass}) = (0.99)^{100} \approx 0.3660$. Let X_i be one when contact i passes. Then $E[\sum_i X_i] = \sum_i E[X_i] = 100(0.99) = 99$. Linearity of expectation does not require independence, but the all-pass product does. A contact test does not establish correct switching, isolation or retention; correlated defects, redundancy and repair alter the whole-array result.

E09

AAO starts with anodization of Al to form pores; material can be deposited into or through the template, possibly after opening the pore bottoms or transferring the membrane. Nanosphere lithography assembles spheres and uses the spheres or interstices as a mask for deposition or etch, followed by mask removal. BCP/DSA forms phase-separated domains whose position is guided by a prepattern; one domain is removed or converted before transfer by etch or lift-off. In every case, check mask defects and transfer fidelity, register features to electrode lines, form contacts and test isolation and electrical addressing. A misplaced feature can miss its contact or bridge a neighbour even if the local pitch looks regular; scanning one site with a conductive probe is not proof that all sites can be addressed by circuitry.